

Introduction:

Seattle is one of the top cities in the country dealing with rapid gentrification, where investment and development often move into areas historically home to low-income residents and communities of color, raising housing costs, attracting higher-income households, and slowly pushing out those already struggling to stay in place. For this project, I focused on low-income households because money plays the biggest role in who gets displaced and who gets prioritized; when a neighborhood is vulnerable, developers can more easily buy property, raise prices, and attract wealthier residents. I also examined light rail stations as a major sign of investment, noticing that most are not placed in the neighborhoods that need transit the most but instead appear in areas already doing well financially, and when they do show up near lower-income areas, they often accelerate gentrification. This project therefore explores how light rail station locations connect to gentrification patterns and highlights the role of low-income communities in understanding who is most at risk of being pushed out. I have seen this firsthand in places like Georgetown, South Park, and Delridge/West Seattle, where one street may hold older, low-income homes while the next block features modern, upscale houses, a stark contrast that illustrates how quickly neighborhoods are changing and who those changes are really for.

Methodology:

For this project, I used 2020 ACS 5-year estimate income data from the U.S. Census Bureau to identify low-income census tracts in Seattle. After importing the table into QGIS, I cleaned the dataset by keeping only the median household income column and joining it to the Seattle census tract shapefile using the GEOID field. I then created my first spatial analysis step: an attribute query selecting tracts with median household incomes below \$70,000. These selected tracts were exported as a new layer for easier mapping. To examine how transit investment relates to gentrification, I downloaded the Seattle neighborhood boundary shapefile and the light rail station point layer from Seattle GeoData. With the light rail stations layer loaded, I performed my second spatial analysis operation by generating a 0.25-mile, 0.5-mile, 0.75-mile, and 1-mile multi-ring buffer around each station. Because some buffers extended beyond the city boundary, I used the Clip tool to keep only the portions within Seattle. After aligning all layers in the same projection, I symbolized the low-income tracts as a graduated choropleth and the buffer rings with soft, transparent colors to highlight how far light rail investment reaches into different neighborhoods. Together, these steps created a clear spatial comparison between where low-income communities are located and how close they are to major transit investments that often accelerate gentrification.

Results:

The analysis shows a clear divide in who benefits from light-rail expansion in Seattle and how it relates to gentrification risk. Mapping median household income across census tracts reveals that higher-income areas cluster along existing light-rail lines, with most stations located in stable or affluent neighborhoods and very few in low-income areas, even though these communities rely most on public transit. Buffer analysis of .25-, .50-, .75-, and 1-mile rings around stations reinforces this pattern: most surrounding tracts are middle-to-upper income, while only a few low-income tracts—such as parts of Rainier Valley, south Beacon Hill, and near the future West Seattle Link—fall within walking distance, often alongside new, upscale development, signaling early displacement risk. Comparing the low-income layer with the buffer layer confirms that low-income tracts are more likely to be outside convenient walking distance, while high-income tracts enjoy shorter distances and better access. This reflects a common trend in gentrifying cities, where transit investments raise property values and attract wealthier residents before benefiting existing communities. Overall, Seattle's light-rail distribution aligns more with existing affluence than with neighborhoods at risk of displacement, meaning low-income communities receive fewer direct benefits while facing greater long-term risk of being priced out as new transit projects arrive.